CCC-C-425b
February 8, 1966
SUPERSEDING
Fed. Spec. CCC-C-425a
September 26, 1961

FEDERAL SPECIFICATION

# CLOTH, BIRDSEYE AND GAUZE; COTTON

*This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.*

## 1. SCOPE AND CLASSIFICATION

**1.1 Scope.** This specification covers the requirements for two types of cotton cloth.

**1.2 Classification.**

**1.2.1** *Types.* The cloth shall be of the following types as specified (see 6.2):

Type I—Birdseye - single layer.
Type II—Gauze - double layer.

## 2. APPLICABLE DOCUMENTS

**2.1 Specifications and standards.** The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

*Federal Specifications:*

CCC-T-191—Textile Test Methods.
PPP-P-51—Packaging, Packing, and Marking of Textile Fabrics (Woolens, Worsteds, Cottons, Silks, and Synthetics).

*Federal Standards:*

Fed. Std. No. 4—Glossary of Fabric Imperfections.
Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo, Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

*Military Standard:*

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

**2.2 Other publication.** The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Rules and Regulations Under the Textile Fiber Products Identification Act.

(Copies may be obtained free of charge from the Federal Trade Commission, Washington, D.C. 20580.)

## 3. REQUIREMENTS

**3.1 Material.**

**3.1.1** *Cotton.* The cotton yarn shall be made from clean cotton and shall be carded, drawn, and spun.



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**3.2 Color.** The cloth shall be bleached white.

**3.3 Physical requirements.** The finished cloth shall conform to the requirements shown in table I or II when tested as specified in 4.4.

TABLE I. *Physical requirements—type I, birdseye*

| Weight, minimum (ounces/ square yard) | Width (inches) (including selvages) | Yarns per inch, minimum | | Breaking strength, minimum (pounds) | |
|---|---|---|---|---|---|
| | | Warp | Filling | Warp | Filling |
| 4.0 | 26-1/2 min.<br>27-1/2 max. | 60 | 48 | 40 | 60 |

TABLE II. *Physical requirements—type II, gauze*

| Weight[1] minimum (ounces/ square yard) | Width (inches) (including selvages) | Yarns per inch, minimum[2] | Breaking strength[1] minimum (pounds) | |
|---|---|---|---|---|
| | | Sum of warp and filling | Warp | Filling |
| 3.35 | 39½ min.<br>40½ max. | 74 | 45 | 25 |

[1] Utilizing two plys of panel.

[2] Determined on single ply of panel, but both single panels shall conform to this requirement.

3.3.1 *Weave.*

3.3.1.1 *Type I.* Type I cloth shall be a single layer cloth in a birdseye weave.

3.3.1.2 *Type II.* Type II cloth shall be a double layer gauze cloth woven in a plain, birdseye, or interlocking-type weave. The combining, or interlocking, of the double layer to form a single layer shall be throughout the length of the cloth, and at regular intervals filling-wise to form pinking bars.

3.3.1.2.1 *Pinking bar.* A filling-wise bar (pinking bar) 1-1/2 ± 1/2 inch wide shall be woven into the cloth at regular intervals so as to form panels of the double layer (2-ply cloth). The 1-1/2 inch pinking bars shall be made by weaving the warps of the 2-ply cloth to form a single layer cloth.

3.3.1.2.2 *Panels.* The panels of 2-ply cloth formed by the pinking bars shall measure 19 ± 1/2 inch in length from inside of pinking bar to inside of pinking bar.

3.3.1.2.3 *Folding stripes.* The supplier may, at his option, insert folding stripes. If folding stripes are used, one stripe consisting of four ends weaving as one in a group of 3 or 4 shall be positioned 13-1/2 ± 1/2 inch from one selvage. Another stripe shall be positioned 8 ± 1/2 inch from the opposite selvage. The folding stripes shall become part of the single layer pinking bars at point of intersection.

**3.4 Shrinkage.** The sum of the warp and filling shrinkage for type I or II cloth shall not exceed 20 percent when tested as specified in 4.4.

**3.5 Absorbency.** Types I and II cloth shall show a minimum absorption of 80 percent when tested as specified in 4.4.

**3.6 Finish.** Types I and II cloth shall be cleaned, and shall be given a full chemical bleach.

3.6.1 *Nonfibrous materials.* The starch and protein content including chloroform-soluble and water-soluble material of the finished types I and II cloth shall not exceed 2.0 percent when tested as specified in 4.4.

**3.7 Length and put-up.** Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards. Each length shall be put up on full width rolls as specified in PPP-P-51 (see 6.5).



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3.7.1 *Pieces.* Pieces of type II cloth shall begin and end with a pinked edge located in the middle of the pinking bar.

3.8 **Fiber identification.** Each roll shall be labeled, ticketed, or invoiced for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.9 **Workmanship.** The finished cloth shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable point levels specified.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Inspection.** Inspection shall be in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.2.1 *Yard-by-yard examination.* The required yardage of each piece shall be examined on the face side only, except that on type II the sides shall be alternated with each piece so that all defects as defined in section I of Fed. Std. No. 4 which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.1.1, with the following exceptions: coarse yarns and jerked-in filling, glossary numbers 2, 3, and 26 will not be scored as defects. Only those slubs and knots which exceed the maximum acceptable limit shown for each shall be scored (see fig. 1). None of the 30 contiguous linear yards, (increments of 1 yard on the measuring device of the inspection machine selected for examination from any one piece within the sample) shall be penalized more than four points. The sample unit for this examination shall be one linear yard. The sample size shall be 750 linear yards, regardless of lot size. The sample yardage shall be examined from 25 pieces taken from 25 containers. An approximately equal number of yards shall be examined in each piece of the sample. The lot shall be unacceptable if the points per hundred square yards exceed 45.00 points. Point computation shall be as follows:

$$\frac{\text{Total points scored in sample size} \times 3600}{\text{Sample size (yards)} \times \text{contracted width of the material (inches)}} = \text{Points per 100 sq. yds.}$$

4.2.1.1 *Demerit points.* Demerit points shall be assigned as follows:

For defects up to 3 inches in any dimension. —one point
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension. —two points
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension. —three points
For defects exceeding 9 inches in any dimension. —four points

Note: The following defects, when present, shall be scored four points for each yard in which they occur:

Selvage curled, folded, rolled, or slack continuously or intermittently throughout the piece.
Overall uncleanliness, objectionable odor.
Width, type I, less than 26-1/2 inches or more than 27-1/2 inches.
Width, type II, less than 39-1/2 inches or more than 40-1/2 inches
Uneven weaving throughout the piece.
Not bleached white.



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Type II only:

Length of panels less than 18-1/2 inches or more than 19-1/2 inches.

Width of pinking bar less than 1 inch or more than 2 inches.

Folding stripes, if present, not in specified locations.

4.2.2 *Examination for length.*

4.2.2.1 *Individual pieces.* During the yard-by-yard examination, each piece in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards below the length marked on the ticket shall be considered a defect with respect to length. For type II, any piece which does not begin or end with a pinked edge in the middle of the pinking bar, shall also be considered a defect. The lot shall be unacceptable if two or more pieces in the sample are defective in respect to length.

4.2.2.2 *Total yardage in sample.* The lot shall be unacceptable if the total of the actual lengths of pieces in the sample is less than the total of the lengths marked on the tickets.

4.2.3 *Examination for noncompliance with the Textile Fiber Products Identification Act.* During the yard-by-yard examination, each piece in the sample shall be examined for conformance to the Textile Fiber Products Identification Act. Each piece not labeled in accordance with this act shall be a defect. The lot shall be unacceptable if two or more of these defects occur.

**4.3 Examination of preparation for delivery requirements.** An examination shall be made in accordance with the provisions of PPP-P-51 to determine that packaging, packing, and marking requirements of section 5 of this specification are complied with.

**4.4 Testing of the end item.** The methods of testing specified in CCC-T-191, wherever applicable, and as listed in table III, shall be followed. Except as otherwise indicated, the physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample unit shall be 3 continuous yards, full width of the finished cloth. The lot shall be unacceptable if one or more sample units fail to meet any test requirements specified. The sample size (number of sample units) shall be in accordance with the following:

| Lot size (yards) | Sample size |
|---|---|
| 800 or less | 2 |
| 801 up to and including 22,000 | 3 |
| 22,001 and over | 5 |

TABLE III. *Tests*

| Characteristic | Requirement paragraph | Test method |
|---|---|---|
| Material: | | |
| Cotton | 3.1.1 | [1] |
| Weave | 3.3.1 | Visual[2] |
| Weight, ounce/ square yard[3] | 3.3 | 5041 |
| Yarns per inch: | | |
| Warp[4] | 3.3 | 5050 |
| Filling[4] | 3.3 | 5050 |
| Breaking strength: | | |
| Warp[5] | 3.3 | 5100 |
| Filling[5] | 3.3 | 5100 |
| Shrinkage in laundering: | | |
| Warp[6] | 3.4 | 5550 |
| Filling[6] | 3.4 | 5550 |
| Absorbency | 3.5 | 5500 |
| Nonfibrous materials | 3.6.1 | 2611 |

[1] The supplier shall submit a certificate of compliance for the characteristic. When required, the certificate shall be accompanied by actual test, inspection or other verifiable quality data.

[2] One determination shall be made from each sample unit and the results reported as "pass" or "fail".

[3] For type II — determination to be made on the two layers of body cloth and should not include folding stripes if present.

[4] For type II — determination to be made on single layer of panel, specimens shall be sampled from both layers and must meet the requirement. The specimens should not include folding stripes if present.

[5] For type II — determination to be made of double layer body cloth (not single ply). All tests must



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be clear of, and must not include the pinking bars or folding stripes, if present.

[6] The specimens of type II cloth shall be a complete diaper, double layer, and shall be cut the width of the cloth through the middle of the respective pinking bars, leaving a pinked edge. The samples shall be dried prior to pressing.

## 5. PREPARATION FOR DELIVERY

For civil agency procurements, the definitions and applications of levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

**5.1 Packaging.** Packaging shall be level A or C as specified (see 6.2 and 6.2.1).

**5.1.1** *Levels A and C.* The cloth, put up on rolls, shall be packaged in accordance with the applicable requirements for class 3 fabrics of PPP-P-51.

**5.2 Packing.** Packing shall be level A, B, or C as specified (see 6.2 and 6.2.1).

**5.2.1** *Levels A, B, and C.* Cloth of one type only shall be packed in accordance with the applicable requirements of PPP-P-51.

**5.3 Marking.** In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of PPP-P-51.

## 6. NOTES

**6.1 Intended use.** The cloth covered by this specification is intended for use in the fabrication of infant diapers.

**6.2 Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type of cloth required (see 1.2).

(*c*) Minimum length, if other than specified (see 3.7).

(*d*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

**6.2.1** In the preparation of contracts or orders, it must be noted that various options, choices, and alternatives, as indicated in PPP-P-51, may be exercised in the preparation for delivery of cloth.

**6.3** Level B packaging and packing should be used by ordering agencies only when shipment is known to be made to a holding supply depot or warehouse from which partial shipments will be made to ultimate consumers.

**6.4 Transportation descriptions.** The transportation descriptions and minimum weights applicable to this commodity are:

*Rail:*

Cotton piece goods, finished, not otherwise indexed by name.
Carload minimum weight—any quantity.

*Motor:*

Cotton piece goods, not otherwise indexed.
Carload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

**6.5** For GSA procurement, rolls shall be 40 yards in length with a tolerance of within plus or minus 2 yards.

**MILITARY CUSTODIANS:**

**Army—GL**
**Navy—SA**
**Air Force—67**

*Review activities:*

**Army—MD**
**Navy—SA**

*User activity:*

**DSA—DM**

*Preparing activity:*

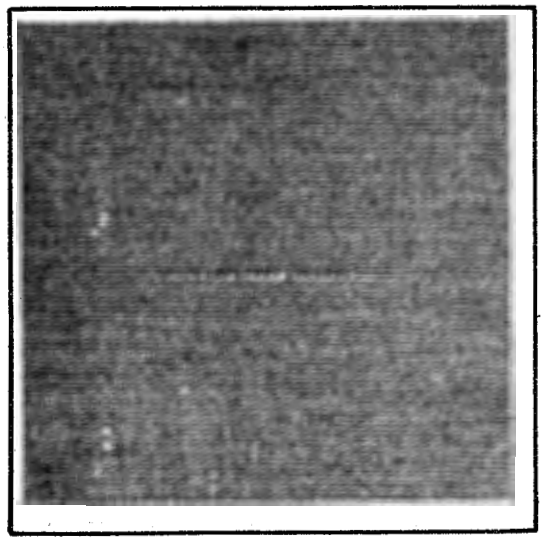
**Army—GL**



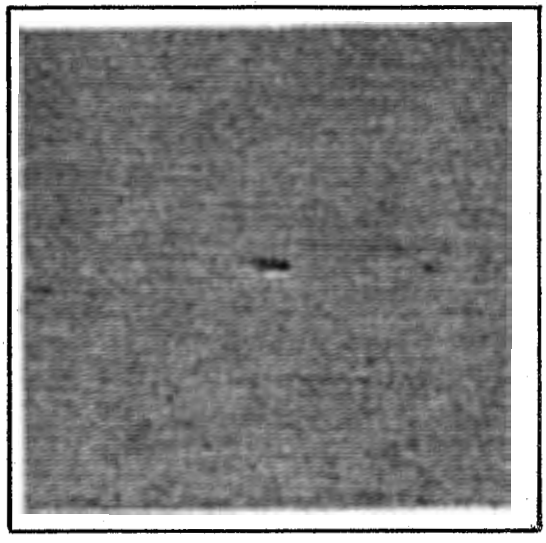
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Slub



Slub (Loose, Thick)



Lump or Knot (Fuzz)

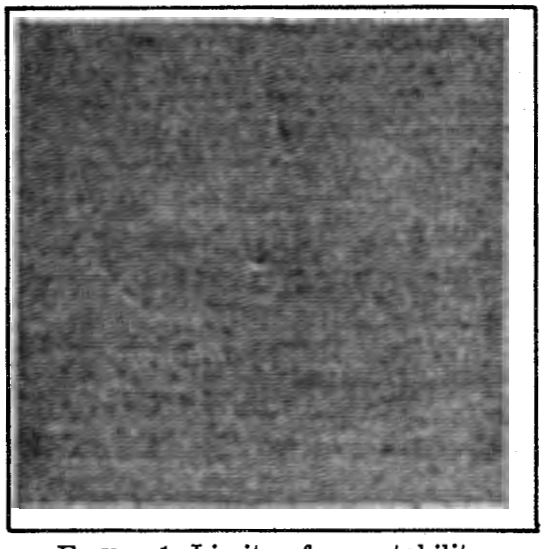


FIGURE 1. Limits of acceptability.

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